Problem: Prove Fermat's Little Theorem, which says that, if $a \in \mathbb{Z}$, and p is a prime number, then $a^p = a \pmod{p}$.

Proof: The set $G = \{1, 2, ..., p-1\}$ forms a group under multiplication modulo p. Let's quickly verify this. We know that multiplication modulo p is associative, because multiplication is associative. We know that 1 is the identity element, because $a \cdot 1 = 1 \cdot a = a \pmod{p}$ for all $a \in G$. Also, we can show that every element $a \in G$ has an inverse. Let a, x, and y be elements of G. Using the rules of modular arithmetic, we know that

$$\begin{aligned} ax = ay \pmod{p} &\implies ax - ay = 0 \pmod{p} \\ &\implies a(x-y) = 0 \pmod{p} \\ &\implies p \mid a \text{ or } p \mid x - y && \text{because } p \text{ is prime} \\ &\implies x = y && \text{because } 0 < a < p \text{ and } 0 < x, y < p \end{aligned}$$

This means that if we fix a and let $X_a = \{ax \mid x \in G\}$, then X_a contains $p-1$ distinct nonzero elements, $X_a = G$, and X_a contains the identity. So there must be some $x \in G$ such that $ax = 1$. Thus every $a \in G$ has an inverse.

Lastly, we can show that G is closed under multiplication modulo p. Every element of G is relatively prime with p, so no product can ever equal 0 modulo p. The products must belong to G, meaning that G is closed.

We have now shown that G is a finite group under multiplication modulo p. Let $a \in G$ and let k be the order of a. By Lagrange's theorem, we know that $k \mid p-1$. The rules of arithmetic allow us to do the following.

$$\begin{aligned} a^k &= 1 \pmod{p} \\ (a^k)^{(p-1)/k} &= 1^{(p-1)/k} \pmod{p} \\ a^{p-1} &= 1 \pmod{p} \\ a^p &= a \pmod{p} \end{aligned}$$

You can see that, once we establish that G is a group under multiplication modulo p, the rest is pretty easy. Lagrange's theorem tells us that $k \mid p-1$, which allows us to raise both sides of the equation to the power of $(p-1)/k$. The k terms cancel and we are left with the equation $a^{p-1} = 1 \pmod{p}$. We multiply both sides by a to obtain the result $a^p = a \pmod{p}$. This holds true for all $a \in G$. We can extend this result to the integers. Let $a \in \mathbb{Z}$ such that $a \neq 0$ and let a' be the element of G such that $a' = a \pmod{p}$. Then $a^p = (a')^p = p \pmod{p}$. Now let $a = 0$. Then $0^p = 0 \pmod{p}$, which shows that it holds for zero as well. Thus it holds for all the integers. QED